STAFF REVIEW · H446 076 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Data are sorted.

2. Extra working memory beyond input storage.

3. A one-time sort has cost.

4. mid4=9, mid6=13, mid7=15; return index7.

I Do — source worked steps

1. Check the working — Linear search: inspect4 then7, returning index1.

We Do — guided CHECK

1. Check the working
mid3=9, mid5=16, mid4=12, return index4.

You Do — source mark schemes

1. Linear [2]; binary requires sorted data [2].

2. mid3=9 [2]; low4 mid5=16 [2]; high4 mid4=12, index4 [2].

3. Linear single search avoids sort [2]; sort once O(n log n) [2], then binary O(log n) repeatedly [2].

Exit Ticket — source answer

1. False. Sorting first may cost more.